



# Day-18 introduction to python with mysql connector

## 1. Introduction to Python with MySQL Connector

Python can be connected with MySQL database using a library called **MySQL Connector**.

It allows Python programs to interact with databases for storing, retrieving, updating, and deleting data.

### ◆ Why use Python with MySQL?

- Python is simple and powerful
- MySQL is fast and widely used database
- Together they are used in:
  - Web applications
  - Student management systems
  - Business applications
  - Data storage systems

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## 2. Key Benefits of Python with MySQL

## ✓ 1. Easy Integration

Python easily connects with MySQL using `mysql-connector-python`.

## ✓ 2. Data Management

You can:

- Insert data
- Fetch data
- Update records
- Delete records

## ✓ 3. Automation

Tasks like:

- Data entry
- Report generation
- Record updates  
can be automated.

## ✓ 4. Scalability

Works for both:

- Small projects (student projects)
- Large applications (companies)

## ✓ 5. Security

- Supports authentication
- Data can be handled securely

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## 3. Installation Steps for MySQL Connector

Follow these steps carefully:

### ◆ Step 1: Install MySQL

- Download and install MySQL Server & MySQL Workbench

### ♦ Step 2: Install Connector in Python

```
pip install mysql-connector-python
```

### ♦ Step 3: Import in Python

```
import mysql.connector
```

### ♦ Step 4: Connect to Database

```
conn=mysql.connector.connect(  
host="localhost",  
user="root",  
password="your_password",  
database="your_database"  
)
```

### ♦ Step 5: Create Cursor

```
cursor=conn.cursor()
```

## 4. Overview of Student Management System

A **Student Management System (SMS)** is an application used to manage student data.

### Main Purpose:

- Store student records
- Manage academic details
- Perform CRUD operations

### ♦ Features of SMS

#### Add Student

```
INSERT INTO students VALUES (...)
```

### ✓ View Students

```
SELECT * FROM students
```

### ✓ Update Student

```
UPDATE students SET name = "... " WHERE id = 1
```

### ✓ Delete Student

```
DELETE FROM students WHERE id = 1
```

## ◆ Example Table Structure

```
CREATE TABLE students (  
    id INT PRIMARY KEY,  
    name VARCHAR(50),  
    age INT,  
    course VARCHAR(50)  
);
```

## 5. Key Takeaways

- Python + MySQL is a powerful combination for backend development
- MySQL Connector allows Python to communicate with databases
- CRUD operations are the core of any database system
- Student Management System is a practical real-world example
- Learning this helps in:
  - Web development
  - Data handling

- Software development
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## 6. Conclusion

Python with MySQL is not just a concept — it is a **real-world skill** used in industry.

If you understand:

- Connection
- Queries
- CRUD

Then you are already building the foundation of **backend development** 

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